Highlights

Eclipse-Modulated Transient Energy Pumping in Near-Polar Low Earth Orbit for Spin-Stabilized Solar Sails

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- Eclipse-modulated SRP can produce bounded transient semi-major-axis gain in near-polar LEO for passive solar sails.
- Gain exists only in a narrow LTDN–pitch region; eclipse causally enables it by suppressing negative-work orbital arcs.
- The effect is self-limiting: apogee pumping, eccentricity growth, and perigee erosion drive eventual breakdown.